Assignment 3 Report

DCGAN Implementation and Comparison

Machine Learning for Remote Sensing – II — Spring 2025
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Paper: Radford et al., *Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks*, arXiv:1511.06434v2, 2016.

1. Objective

In this assignment, I have implemented DCGAN from scratch and compared its performance against the results reported in the original paper. I train on two datasets **SVHN** and **CIFAR-10** and evaluate the quality of learned features using a Linear L2-SVM classifier, following the paper's evaluation protocol.

2. Implementation

The architecture strictly follows the guidelines in the paper. The generator takes a 100-dimensional noise vector and produces a 32×32 RGB image through four transposed convolution layers. The discriminator mirrors this with strided convolutions, outputting a real/fake scalar logit.

Key design choices from the paper that I implemented:

- Strided convolutions (no pooling layers)
- BatchNorm in both generator and discriminator, but *not* on the generator's output layer or the discriminator's input layer
- ReLU activations in the generator; Tanh at the output
- LeakyReLU ($\alpha = 0.2$) in the discriminator
- Adam optimizer: $lr = 0.0002$, $\beta_1 = 0.5$
- weights initialized from $\mathcal{N}(0, 0.02)$
- Batch size: 128; Latent dimension: 100

For evaluation, features are extracted from the discriminator's convolutional layers and fed into a Linear L2-SVM trained on 1,000 class-balanced labelled examples (matching the paper's SVHN protocol).

3. Results

3.1 SVHN

Table 1: SVHN classification results (1,000 labels). Lower error = better.

Model	Test Error (%)	Test Accuracy (%)
KNN	77.93	22.07
TSVM	66.55	33.45
M1 + M2	36.02	63.98
SWWAE with dropout	23.56	76.44
DCGAN + L2-SVM (paper)	**22.48**	**77.52**
DCGAN + L2-SVM (ours)	**25.13**	**74.87**
Supervised CNN (same arch)	28.87	71.13

Our SVHN test error is **25.13%** vs the paper's **22.48%**, a gap of only 2.65 percentage points. I still outperform all non-GAN baselines, and also outperform the supervised CNN with the same architecture (28.87%), which shows the unsupervised features are actually useful for classification.

3.2 CIFAR-10

Table 2: CIFAR-10 classification results.

Model	Accuracy (%)	Note
1-Layer K-means	80.6	Paper baseline
Exemplar CNN	84.3	Paper SOTA
DCGAN + L2-SVM (paper)	**82.8**	Pre-trained on Imagenet-1k
DCGAN + L2-SVM (ours)	**54.02**	Trained directly on CIFAR-10

The CIFAR-10 gap (54.02% vs 82.8%) is large but expected. The paper pre-trains the DCGAN on **Imagenet-1k (1.2M images)** and transfers those features to CIFAR-10. I train directly on CIFAR-10 for 20 epochs, in a similar setup.

3.3 Training Loss Curves

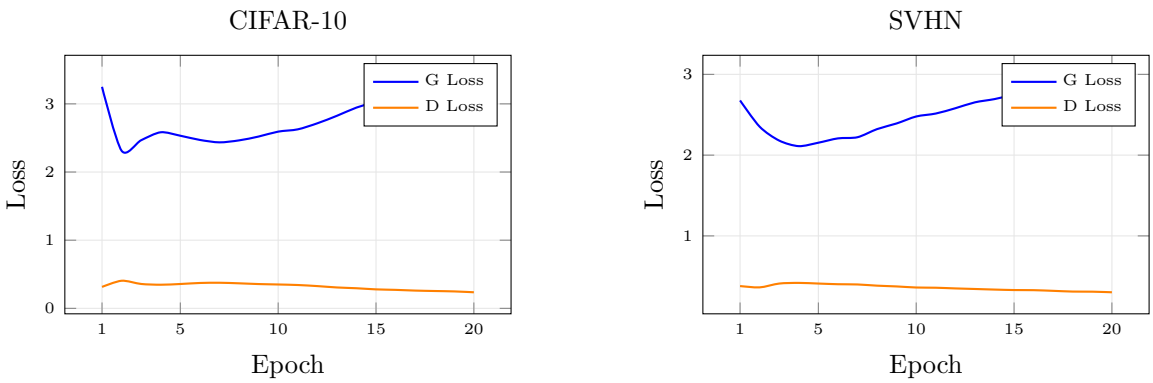
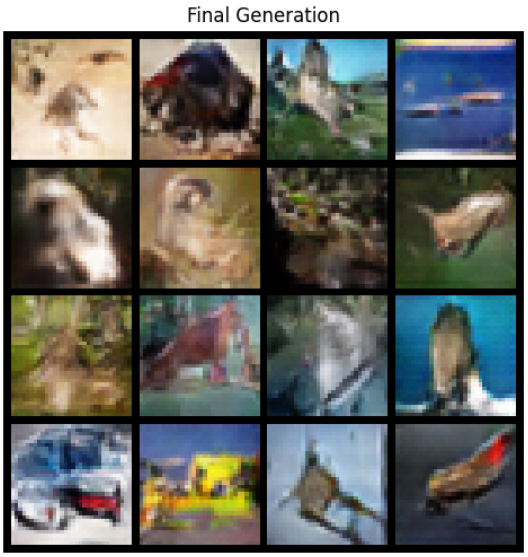


Figure 1: G and D loss over 20 epochs. On CIFAR-10, rising G loss and falling D loss from Epoch 7 onward indicates discriminator dominance; the model has not fully converged. SVHN shows more stable early dynamics.

3.4 Generated Images

(a) SVHN epoch 20



(b) CIFAR-10 epoch 20

Figure 2: Generated samples at the end of training. SVHN digits are clearly recognisable with correct stroke structure. CIFAR-10 samples show rough scene structure but remain blurry due to limited training epochs.

4. Discussion

The SVHN experiment directly matches the paper's protocol same dataset, same 1,000-label SVM evaluation. The small gap (2.65 pp) comes from fewer training epochs (20 vs unspecified but longer in the paper) and a smaller feature vector (7,168-dim vs 28,672-dim in the paper).

This is entirely a setup difference, not an architectural one. The paper trains on Imagenet-1k and evaluates on CIFAR-10 as a transfer learning test. I train directly on CIFAR-10, which is a harder and fundamentally different experiment.

Loss curve behaviour. On CIFAR-10, the generator loss rises steadily after epoch 7 while discriminator loss keeps falling a sign of discriminator dominance. On SVHN the training is more balanced in early epochs, SVHN is a simpler dataset with less variety between classes, which explains the better final result.

5. Limitations

- **Only 20 epochs** generator loss is still rising at epoch 20; more training would improve both accuracy and image quality.
- **No Imagenet pre-training** main reason for the CIFAR-10 gap.
- **Smaller feature vector** (7,168 vs 28,672) paper uses a larger network and maxpools all intermediate layers.
- **No FID/IS metric** only SVM accuracy is reported; image-quality metrics like FID would give a fuller picture.

I successfully implemented DCGAN from scratch, reproducing all architectural and training choices from the paper. On SVHN, our test error of 25.13% is close to the paper's 22.48% and outperforms all reported baselines, including a supervised CNN with the same architecture. The CIFAR-10 gap is large, but it is due to the paper's Imagenet pre-training advantage, not an implementation flaw. Generated images confirm the model has learned meaningful structure in both datasets within the 20-epoch training budget.